

Nickole Andrea Villabona

Irvine, California.

+1 (949) 2797207 • nvillabo@uci.edu

Research Interests

Microbial ecology, evolutionary ecology, microbial evolution, genomics, biology of symbiosis, tropical ecology, bees, bioinformatics.

Education

4th year Ph.D. Candidate in Ecology and Evolutionary Biology	Fall 2022 - Present
<i>University of California, Irvine, USA</i>	

Major in Biology

<i>Universidad de los Andes, Bogotá, Colombia</i>	May 2022
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Major in Microbiology

<i>Universidad de los Andes, Bogotá, Colombia</i>	May 2022
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Minor in Narrative Media

<i>Universidad de los Andes, Bogotá, Colombia</i>	May 2022
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Peer-Reviewed Publications

Villabona, N., Moran, N., Hammer, T., & Reyes, A. (2023). Conserved, yet disruption-prone, gut microbiomes in neotropical bumblebees. *mSphere*, e0013923. <https://doi.org/10.1128/msphere.00139-23>

Mahecha, L., **Villabona, N.**, Sierra, L., Ocampo, D., & Laverde-R, O. (2018). The Andean Cock-of-the-rock (*Rupicola peruvianus*) is a frugivorous bird predator. *The Wilson Journal of Ornithology*, 130(2), 558-560.

Pre-prints (Not peer reviewed yet)

Villabona Buitrago, N. A., Vazquez, A., Schlauch, J., Rasmussen, C., Joyce, R., & Hammer, T. J. (2026). Gain and loss of gut symbionts during stingless bee diversification is linked to host group size. bioRxiv 2026.01.16.699994; doi: <https://doi.org/10.64898/2026.01.16.699994>

Research Experience

Ph.D. Candidate. Hammer Lab. Department of Ecology and Evolutionary Biology. School of Biological Sciences. University of California, Irvine.	Fall 2020 – Present
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Summer Undergraduate Program for Experiential Research	Jun 2019 – Aug 2019
<i>University of Texas at Austin, Austin, Texas</i>	

- Worked under the guidance of Dr. Tobin Hammer and Dr. Nancy Moran. Practiced molecular laboratory techniques under an independent project aimed at sequencing bumblebee gut microbiomes from specimens I collected in the Neotropics.

Undergraduate Research Assistant

<i>Universidad de Los Andes, Bogotá, Colombia</i>	Oct 2018 – Present
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- Research under the guidance of Dr. Alejandro Reyes in the Computational Biology and Microbial Ecology laboratory (BCEM). Learned bioinformatics and microbial ecology techniques. Also had the opportunity to share knowledge and experience with Master's and Ph.D. students.

Undergraduate Research Assistant Feb 2018 – Jan 2020

Universidad de los Andes, Bogotá, Colombia

- Research under the guidance of Dr. Emilio Realpe in the Zoology and Aquatic Ecology laboratory (LAZOE). Learned about arthropod phylogenetics, ecology and evolution. Also had the opportunity to share knowledge and experience with Master's and Ph.D. students.

Professional Experience

Marketing and social media managing.

- Biome Acuity - microbiome service company **Apr 2022 – Oct 2022**

Research consultant in microbial ecology and bioinformatics

- CorpoGen - Research and Biotechnology **Jul 2021 – Aug 2022**

Organizing activities for conferences, meetings, etc.

- Organizing Committee Member, ISME Latin American Congress, Bogotá, July 2021.
- Instructor of QIIME2 Workshop, ISME Latin American Congress, Bogotá, July 2021.
- Writer, editor, and translator for eTutorial at CABANA. Bogotá. March 2022.

Research

- Research assistant Vice-chancellor office for research, Universidad de Los Andes
Sep 2019 – July 2020
- Graduate Student Researcher, Department of Ecology and Evolutionary Biology, University of California, in Irvine.
Jan 2023 – Mar 2023
- Graduate Student Researcher, Department of Ecology and Evolutionary Biology, University of California, in Irvine.
Jul 2023 – Sept 2023

Teaching Assistant

- Principles of Ecology, Department of biological sciences, Universidad de Los Andes, Bogotá, Colombia.
Aug 2019- Dec 2019
- Invertebrate Biology, Department of biological sciences, Universidad de Los Andes, Bogotá, Colombia.
Jan 2018 – May 2019
- Teaching Assistant, Department of Ecology and Evolutionary Biology, University of California, in Irvine.
Apr 2023 – Jun 2023
- Teaching Assistant, Department of Ecology and Evolutionary Biology, University of California, in Irvine.
Oct 2023 – Winter 2026

Presentations

2025. Winter Ecology and Evolutionary Biology Graduate Symposium. Irvine, CA, United States.

Group size mediates maintenance of specialized gut symbionts across stingless bee diversification. *Poster presentation.*

2024. SoCal Microbiome Symposium. Irvine, CA, United States. Group size mediates maintenance of specialized gut symbionts across stingless bee diversification. *Poster presentation.*

2024. ISME19 International Symposium on Microbial Ecology. Cape town, South Africa. Group size mediates maintenance of specialized gut symbionts across stingless bee diversification. *Poster presentation, ISME Early Career Researcher Poster Award*

2023. XII Congreso Mesoamericano de Abejas nativas. Ciudad Guzman, Mexico. Abejorros del Neotrópico presentan una microbiota conservada, pero también propensa a alteraciones. *Oral presentation*

2023. Yosemite Symbiosis Workshop XI. Yosemite national Park, United States. Neotropical bumblebee gut Microbiota are host conserved, yet disruption prone. *Oral presentation*

2022. Entomology 2022 Joint Annual Meeting. Vancouver, Canada. Neotropical bumblebee gut Microbiota are host conserved, yet disruption prone. *Oral presentation*

2022. Sixth Congress of Computational Biology and Bioinformatics. Cartagena, Colombia. La microbiota intestinal de abejorros del neotrópico es conservada, pero propensa a disrupción. ***Oral presentation, Awarded Best Talk***

2021. ISME Latin American Congress 2021. Bogotá, Colombia. A. First exhaustive gut microbiota characterization of neotropical bees. *Online oral presentation.*

2020. Biological Sciences Conference, Universidad de los Andes. Bogotá, Colombia. A. First exhaustive gut microbiota characterization of neotropical bees. *Online oral presentation*

2019. Fourth International Forum on Computational Biology, Bogotá, Colombia. ¿Cómo varía la microbiota intestinal en Abejorros del Neotrópico (Bombus sp.) en distintas altitudes? *Poster presentation*

2019. Fifth Conference on Computational Biology and Bioinformatics in Colombia. Ibagué, Colombia. ¿Varía la microbiota intestinal en Abejorros del Neotrópico (Bombus sp.) en distintas altitudes? *Oral presentation*

2019. Research symposium for the UT SUPER program, Austin, Texas, US. Does neotropical bumblebee (Bombus) gut microbiota vary between different altitudes? *Oral and poster presentation*

Grants and fellowships

2024 Miguel Velez Scholarship. University of California in Irvine.

Workshops

2025. Workshop on Molecular Evolution, The University of Chicago, Marine Biological Laboratory.

2023. Introduction to the Meliponiculture. Ciudad Guzman, Mexico.

Skills

Molecular Biology Techniques

- DNA extraction from tissue, blood, fecal, and hair samples
- PCR, qPCR and gel electrophoresis
- Amplicon and genomic library preparation (Ion Torrent)
- DNA quantification (Qubit, Bioanalyzer)
- Illumina-compatible amplicon library preparation

Classical Microbiology & Laboratory Training

- BSL-2 laboratory practices (supervised coursework)
- Bacterial isolation and culturing; aseptic technique
- Solid and liquid media preparation
- Biochemical assays for bacterial identification
- Antibiotic susceptibility testing
- Phage inoculation assays

Computational & Bioinformatics Skills

- Genomic and metagenomic data analysis
- Phylogenetic and evolutionary analysis
- Statistical analysis and visualization in R
- Bash scripting and Unix/Linux command-line workflows

Scientific Communication & Visualization

- Figure and presentation design for scientific articles
- Adobe Illustrator, BioRender, Photoshop, Premiere; Reaper.

Field Biology & Organismal Skills

- Insect sampling (active and passive methods)
- Insect identification and dissection
- Specimen preservation and pinning

In-depth courses

Master's level courses: Arthropod phylogenetics, Advanced molecular biology, Scientific Communication, Microbial ecology and bioinformatics.

Undergraduate courses: Ecology, Invertebrates Biology, Vertebrates Biology, Entomology, Limnology, Bacteriology, Fungal biology, Genetics, Ecology, Development, Plant Physiology, Forensic genetics, Evolution, Conservation biology, Virology, Statistical inference and bioinformatics, Molecular biology, Genomics and Bioinformatics.

Graduate courses: Quantitative methods, Evolution, Ecology and Phylogenetic Comparative Methods.

Leadership and Volunteering

- Created and led: Workshop in microbial ecology for undergraduate students **Winter 2026 – current**

-Led "BEEMAS" discussion group for graduate and undergraduate students at La Universidad de Los Andes.

- Generate and share knowledge of bees in a diverse environment.

Jan 2020 – Dec 2020

-Volunteer for “Progresía Fenicia”.

- Social work with children from low-income settlements and exposed to insecure environments to strengthen their development within community spaces through academic support and playful activities.

Jan 2017 – Dec 2017

Languages

- Native Spanish
- English B2 - TOEFL certified

Academic references

Dr. Tobin Hammer. Department of Ecology and Evolutionary Biology. University of California, Irvine, Irvine, CA 92697. E-mail: hammert@uci.edu

Dr. Jenniffer Martiny. Department of Ecology and Evolutionary Biology. University of California, Irvine, Irvine, CA 92697. E-mail: jmartiny@uci.edu

Dr. Alejandro Reyes. Departamento de Ciencias Biológicas, Universidad de los Andes. Bogotá, Colombia. Email: a.reyes@uniandes.edu.co